A dimension bound of eleven for right-angled hyperbolic polytopes

Coxeter Polytopes Frontier Project

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Abstract

We exclude finite-volume right-angled hyperbolic polytopes with both finite and ideal vertices in dimension twelve. A canonical three-face defines an undirected opposition graph on the finite vertices. Its invariant Lorentz vector allows one fixed null direction on each component. Exact subset moments and a positive-part quartic majorant give $2B \le (1420561/12500)f$, where B counts these three-faces and f counts finite vertices. Polygon incidences and an exception budget for three-faces give the incompatible strict lower bound $2B > (184460/1619)f$. Together with the published higher-dimensional exclusion, the mixed dimension bound is eleven. The compact and entirely ideal cases have smaller published bounds, so eleven also bounds all finite-volume right-angled polytopes. The geometric and scalar arguments are given in full. A small exact coefficient certificate and conditional Lean checks of the final counting logic are separate verification layers; the geometry is not fully formalized, and no new example or sharpness claim is made.

1 Statement, conventions and external inputs

Theorem 1.1. *A finite-volume right-angled hyperbolic polytope having both finite and ideal vertices has dimension at most eleven.*

Corollary 1.2. *Every finite-volume right-angled hyperbolic polytope has dimension at most eleven.*

We use Dufour's published bound twelve [1] to reduce Theorem 1.1 to dimension twelve. The compact and entirely ideal bounds four and six [2, 3] then give Corollary 1.2. These previous bounds are theorem inputs, not conclusions of the new opposition argument. Examples in dimension eight do not establish sharpness of the new upper bound.

Until the final contradiction let $P \subset \mathbb{H}^{12}$ be a hypothetical mixed right-angled polytope. Write $f > 0$ for its finite vertices, $c > 0$ for its ideal vertices, and a_j for its j-faces. The letter B will count a specific three-face defined below; it never denotes the number of triangular polygons or all three-faces.

Use Lorentz signature $(12, 1)$, future unit timelike representatives $q^2 = -1$ for finite points, and outward unit spacelike normals r. Wall inequalities are $\langle x, r\rangle \le 0$. Distinct outward normals have nonpositive products. At a finite vertex q, the twelve incident normals $n_1, \dots, n_{12}$ are orthonormal; direction i is the edge obtained by omitting n_i. Ideal vertices are future null rays. Faces of dimension at least three inherit finite volume and right angles. Two-faces are hyperbolic polygons with finite angles $\pi/2$ and ideal angles zero.

The external structural inputs are these standard acute-wall/link facts, simple truncation at ideal vertices and Dehn–Sommerville symmetry, and the published lower bound of 82 for the number of facets of a finite-volume right-angled eight-polytope. The last bound and the truncation conventions are recorded in Alexandrov's Sections 2–6 [4]. The absence of compact

right-angled polytopes above dimension four ensures that this noncompact lower bound applies to every eight-face used below. All small-three-face geometry, moment calculations, scalar inequalities and incidence eliminations needed for the new exclusion are proved here. No restriction on a facet spectrum or arithmetic field is imposed.

2 Actual three-face oppositions

2.1 Projection and the normal-product gap

Lemma 2.1. *Two distinct unit outward facet normals of a right-angled polytope have scalar product zero or at most* -1.

Proof. Acuteness gives a product $g \le 0$. Suppose $-1 < g < 0$. Their Gram matrix $G = \left(\begin{smallmatrix} 1 & g \\ g & 1 \end{smallmatrix}\right)$ is positive definite and G^{-1} is entrywise nonnegative. For a strict future timelike interior point x, put $b = (\langle x, u\rangle, \langle x, v\rangle)^T < 0$ and $x_0 = x - [u\ v]G^{-1}b$. The coefficients added to x are positive, so every other wall r satisfies $\langle x_0, r\rangle \le \langle x, r\rangle < 0$. Moreover $x_0^2 = x^2 - b^T G^{-1} b < 0$ and $\langle x_0, x\rangle < 0$. Thus x_0 is a relative-interior point of an actual ridge with a nonright angle, a contradiction. □

The same argument with an orthonormal root set proves actual feasibility of its projected intersection. Positive null sums of actual outward normals are also useful: acuteness checks every other wall, and strict interior pairing proves such a sum is nonzero and future. It therefore represents a feasible ideal point.

Call a two-face with one finite and two ideal vertices a *bad triangle*; the name denotes a counted type, not a forbidden face. At a finite q, form a graph G_q on the twelve directions, with ij an edge when their two-face is a bad triangle. Its two ideal endpoints are $q - n_i, q - n_j$. The opposite wall has the uniquely oriented unit normal

$$m_{ij} = q - n_i - n_j. \tag{1}$$

This follows by orthogonality to the other incident normals, incidence with the two endpoints and unit norm; the displayed sign is strict at q.

Lemma 2.2. *Each triangle* $S = \{i, j, k\}$ *in* G_q *determines an actual three-face* P_3*, the triangular bipyramid with two finite vertices and three ideal vertices. Its other finite vertex is*

$$q_S = 2q - n_i - n_j - n_k. \tag{2}$$

Its six polygon facets are bad triangles.

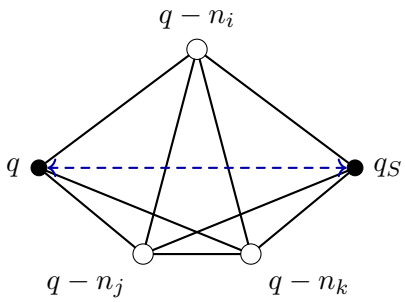
Proof. The three actual normals m_{ij}, m_{ik}, m_{jk} are orthonormal. The displayed q_S has square -1 and $\langle q, q_S\rangle = -2$, so is future timelike. It vanishes on these three walls and on the n_l outside S. The old n_i, n_j, n_k have strictly negative products there.

To check all remaining walls, write their unit normals in the complete q, n_l frame as $r = aq - \sum_l c_l n_l$. Strictness at q gives $a > 0$ and acuteness gives $c_l \ge 0$. For r distinct from the three selected m walls, their acute products imply

$$c_i + c_j \le a, \qquad c_i + c_k \le a, \qquad c_j + c_k \le a.$$

Consequently $\langle r, q_S\rangle = -2a + c_i + c_j + c_k \le -a/2 < 0$. Thus q_S is an actual simple finite vertex with exactly the specified incident walls.

In the Lorentz four-space spanned by q, n_i, n_j, n_k, the six known inequalities are, for $x = tq - \sum y_l n_l$, $y_l \ge 0$ and $y_l + y_m \le t$. They enclose precisely the finite-volume convex hull of $q, q_S, q - n_i, q - n_j, q - n_k$. All five points belong to P. Convexity therefore prevents any additional wall from cutting this hull; the three-face is exactly that bipyramid. Its six facets have the asserted vertex types. □



$q_S = 2q - n_i - n_j - n_k$
two finite vertices, one undirected opposition

Figure 1: The actual P_3 three-face. This is an incidence drawing, not a metric projection. Hollow vertices are ideal. The dashed opposition segment is not an edge of the polytope.

Let B count all three-faces isometric to P_3. Every such face gives one triangle of G_q at each of its two finite vertices, and Lemma 2.2 gives the converse. Different triples at q give different opposites, by linear independence of the n_i. Thus they form a finite undirected graph on all finite vertices, with exactly B edges and degree sum $2B$.

2.2 A fixed null direction on each opposition component

Lemma 2.3. *The vector $h = 3q - \sum_{i=1}^{12} n_i$ is constant across every actual opposition edge. It has $h^2 = 3$ and $\langle h, q\rangle = -3$. On each opposition component one can fix a nonzero future null vector Z with $\langle Z, h\rangle = 0$.*

Proof. At q_S the new three normals are the m in (1); the other nine normals are unchanged. Their total is $3q + \sum_i n_i - 3\sum_{i\in S} n_i$. Substitution in $3q_S$ minus this sum gives exactly h. The stated square and product follow from the orthonormal frame. Since h is spacelike, its orthogonal complement is Lorentzian and contains a future null direction. Choose one on each component, including isolated vertices. The choice does not require complete corner graphs. □

Put

$$E(q) = -\langle Z, q\rangle > 0, \qquad p_i = -\frac{\langle Z, n_i\rangle}{E(q)}, \qquad g(q) = E(q)^{-2}. \tag{3}$$

The same Z is used at both endpoints of every actual edge. Componentwise rescaling of Z cancels from all edge ratios. Decomposition $Z = E(q)(q - \sum p_i n_i)$, together with $Z^2 = 0$ and $\langle Z, h\rangle = 0$, gives

$$\sum_i p_i = 3, \qquad \sum_i p_i^2 = 1. \tag{4}$$

There is no assumption that individual p_i are nonnegative.

3 Formal triples and their exact moments

Set $x_i = p_i - 1/4$. Then $\sum x_i = 0$ and $\sum x_i^2 = 1/4$. For *all* 220 three-subsets S define the formal vector $q_S = 2q - \sum_{i\in S} n_i$, whether or not it is an actual opposition. It is future unit timelike, since $q_S^2 = -1$ and $\langle q, q_S\rangle = -2$. Its formal energy ratio is therefore positive:

$$r_S = \frac{E(q_S)}{E(q)} = 2 - \sum_{i\in S} p_i = \frac{5}{4} + d_S, \qquad d_S = -\sum_{i\in S} x_i. \tag{5}$$

The centered indicator of a three-subset has squared norm $9/4$. Cauchy–Schwarz gives $|d_S| \le 3/4$, hence

$$\frac{1}{2} \le r_S \le 2. \tag{6}$$

No formal vector is presumed to be a feasible chamber vertex.

Lemma 3.1. *Uniform averages over the* 220 *formal triples at a fixed finite vertex satisfy*

$$\mathbb{E}d = 0, \qquad \mathbb{E}d^2 = \frac{9}{176},$$
$$\mathbb{E}d^3 = -\frac{27}{220}\sum_i x_i^3, \qquad \mathbb{E}d^4 = \frac{3}{440} - \frac{3}{220}\sum_i x_i^4 \le \frac{3}{440}.$$

Moreover $-\mathbb{E}d^3 \le 27/1760$.

Proof. The inclusion probabilities for one, two and three specified indices are respectively $1/4, 1/22, 1/220$. With $s_j = \sum_i x_i^j$ and $s_1 = 0$, the second moment is $(1/4 - 1/22)s_2 = 9/176$. The coefficient of s_3 in the third subset-sum moment is $1/4 - 3/22 + 2/220 = 27/220$; the sign reverses for d. The fourth expansion is

$$(1/4 - 7/22 + 12/220)s_4 + (3/22 - 6/220)s_2^2 = -3s_4/220 + 3/440.$$

Finally $|s_3| \le \sum |x_i|^3 \le s_2^{3/2} = 1/8$. □

4 A positive-part quartic bound

The pointwise scalar inequality needed for deletion of absent formal triples is stronger than a reciprocal estimate alone.

Lemma 4.1. *For* $-3/4 \le d \le 3/4$, *put*

$$P(d) = \frac{19}{125} - \frac{979}{1000}d + \frac{768}{625}d^2 - \frac{971}{3125}d^3 + \frac{53186}{9375}d^4.$$

Then

$$P(d) \ge \max\left\{0, \left(\frac{5}{4} + d\right)^{-2} - \frac{1}{2}\right\}. \tag{7}$$

Proof. We prove nonnegativity and reciprocal majorization separately. For $d \le 0$ every term of P is nonnegative. For $d \ge 0$,

$$P''(d) - \frac{12}{5} = \frac{36}{625} - \frac{5826}{3125}d + \frac{212744}{3125}d^2 \ge \frac{36}{625} - 2d + 60d^2 > 0.$$

The last quadratic has minimum $36/625 - 1/60 > 0$. At $d_0 = 1/4$,

$$P(d_0) = \frac{1627}{1200000} > \frac{13}{10000}, \qquad P'(d_0) = -\frac{10243}{150000}, \quad |P'(d_0)| < \frac{7}{100}.$$

The strong-convexity lower parabola gives

$$P(d) \ge \frac{13}{10000} - \frac{(7/100)^2}{24/5} = \frac{67}{240000} > 0 \qquad (d \ge 0).$$

Write $r = \mu + d$, $\mu = 5/4$. The exact reciprocal-square identity is

$$r^{-2} = \mu^{-2} - 2d\mu^{-3} + 3d^2\mu^{-4} - 4d^3\mu^{-5} + \frac{d^4(4r + \mu)}{\mu^5 r^2}. \tag{8}$$

It follows by multiplying by $\mu^5 r^2$ and expanding. Since $(4r+\mu)/r^2$ decreases for $r>0$, its maximum for $r \geq 1/2$ is 13. Thus $r^{-2} - 1/2$ is at most

$$P_0(d) = \frac{7}{50} - \frac{128}{125}d + \frac{768}{625}d^2 - \frac{4096}{3125}d^3 + \frac{13312}{3125}d^4.$$

For $d \geq 0$ the difference

$$P - P_0 = \frac{3}{250} + \frac{9}{200}d + d^3 + \frac{106}{75}d^4$$

is nonnegative. This proves reciprocal majorization on that half.

For $d = -t$, $0 \leq t \leq 3/4$, direct multiplication gives

$$\begin{aligned} N(t) &= \left(\frac{5}{4} - t\right)^2 \left(P(-t) - \left(\frac{5}{4} - t\right)^{-2} + \frac{1}{2}\right) \\ &= \frac{3}{160} - \frac{321}{3200}t + \frac{249}{2000}t^2 - \frac{643}{400}t^3 + \frac{27949}{3000}t^4 - \frac{130052}{9375}t^5 + \frac{53186}{9375}t^6. \end{aligned} \tag{9}$$

We give a rational certificate that $N > 0$ on this entire interval. For a degree-m polynomial Q, write $Q(a+(b-a)u) = \sum_{k=0}^{m} c_k u^k$ and define

$$\beta_i = \sum_{k=0}^{i} c_k \frac{\binom{i}{k}}{\binom{m}{k}}.$$

The binomial theorem gives $Q(a+(b-a)u) = \sum_i \beta_i \binom{m}{i} u^i (1-u)^{m-i}$. On $0 \leq u \leq 1$ the basis is nonnegative and sums to one. Thus bounds on these coefficients control the full interval. Expansion of (9) gives the following strict bounds on all successive coefficients; the displayed formula makes each a direct rational check.

Polynomial and interval	Bounds on successive Bernstein coefficients
N' on $[0, 1/5]$, degree 5	Less than $-1/10, -9/100, -99/1000, -98/1000, -79/1000, -9/200$
N' on $[3/10, 1/2]$, degree 5	Greater than $3/50, 11/100, 17/100, 21/100, 23/100, 1/5$
N'' on $[1/2, 3/4]$, degree 4	Less than $-3/5, -19/10, -37/10, -29/5, -36/5$

The first row makes N decreasing on $[0, 1/5]$, where $N(1/5) > 1/1000$ proves positivity.

On the central interval $[1/5, 3/10]$, the quadratic N'''' is convex and has endpoint values less than $-27, -91$, so it is negative throughout. Hence N''' decreases; its endpoint value $N'''(3/10) > 9/10$ makes it positive. Therefore N'' increases and $N''(1/5) > 4/5$ makes $N'' > 1/2$ throughout. The exact midpoint values are

$$N(1/4) = \frac{679}{1200000} > \frac{1}{2000}, \qquad N'(1/4) = \frac{3061}{600000}, \quad |N'(1/4)| < \frac{1}{100}.$$

Its lower parabola gives $N \geq 1/2000 - (1/100)^2 > 0$ there. The second Bernstein row makes N increasing on $[3/10, 1/2]$, so positivity propagates from the central endpoint. The third makes N concave on $[1/2, 3/4]$. Its first endpoint is positive by the preceding interval and its last is $N(3/4) = 57/64000 > 0$. A concave function lies above its chord. These four intervals cover $[0, 3/4]$, proving $N > 0$ and hence the remaining reciprocal bound. □

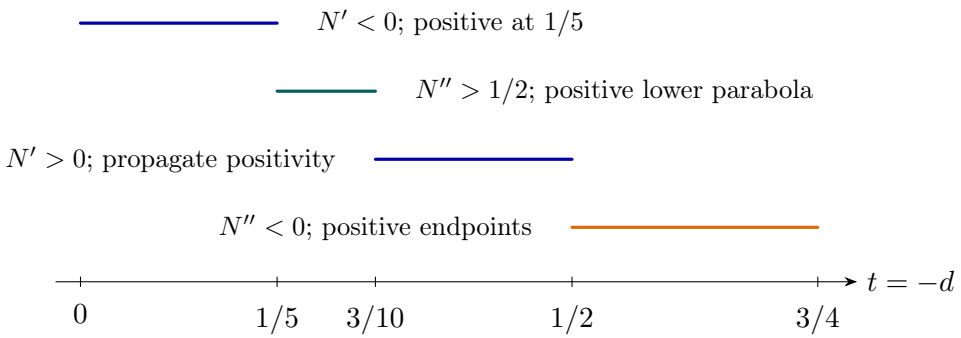


Figure 2: The exact interval proof for N. The bars indicate certificate domains, not sampled values or a plotted approximation to the polynomial.

5 Deleting absent triples and summing ordinary vertices

Proposition 5.1. *Every finite-volume right-angled twelve-polytope satisfies*

$$2B \le \frac{1420561}{12500} f. \tag{10}$$

Proof. At a finite vertex let $t(q)$ be its opposition degree. For each actual neighbor, equations (3)–(5) give $g(q_S)/g(q) = r_S^{-2}$. Lemma 4.1 implies

$$r_S^{-2} - 1 \le P(d_S) - 1/2.$$

Sum over the actual subset of triples. Inserting the absent formal triples on the right is legitimate because $P \ge 0$, giving

$$\sum_{\text{actual } S} \left(\frac{g(q_S)}{g(q)} - 1 \right) \le \sum_{\text{all 220 } S} P(d_S) - \frac{t(q)}{2}. \tag{11}$$

The cubic coefficient of P is negative, and its quadratic and quartic coefficients are positive. Lemma 3.1 therefore gives, pointwise at every finite vertex,

$$\mathbb{E}P \le \frac{19}{125} + \frac{768}{625}\frac{9}{176} + \frac{971}{3125}\frac{27}{1760} + \frac{53186}{9375}\frac{3}{440} = \frac{1420561}{5500000}. \tag{12}$$

Now sum (11) over *ordinary unweighted* finite vertices, allowing the componentwise choices of Z. Each actual undirected edge contributes a reciprocal pair

$$\frac{g(v)}{g(u)} + \frac{g(u)}{g(v)} - 2 = r^{-2} + r^2 - 2 \ge 0.$$

Using $\sum_q t(q) = 2B$ gives $0 \le 220(1420561/5500000)f - B$, which is (10). This summation does not select a height maximum or turn a weighted average into an unweighted one. The constant $1/2$ is a deletion allowance in the scalar bound, not an added move or multiplicity. □

6 A strict weighted polygon row

We next derive the opposing lower bound for the *same* B. At a finite vertex there are $\binom{12}{2} = 66$ two-faces. At an ideal vertex the cubical eleven-dimensional link has 11264 edges, so there are 11264 two-faces. With $\lambda = 2123/1280$, we need

$$66f + 11264\lambda c < \frac{22}{5} a_2, \qquad \text{equivalently } a_2 > 15f + 4246c. \tag{13}$$

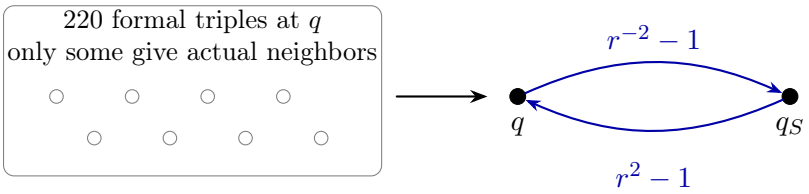


Figure 3: The actual-subset passage and reciprocal edge pairing. The left panel is a schematic row, not a claim that every formal triple is a chamber face. Summation on the right uses the actual graph only.

Here is an explicit Dehn–Sommerville derivation. Truncate every ideal vertex. The simple Euclidean twelve-polytope so obtained has face counts

$$b_0 = f + 2048c, \quad b_{11} = a_{11} + c, \quad b_k = a_k + 2^{11-k}\binom{11}{k}c \quad (1 \leq k \leq 10).$$

In particular b_0 replaces each old cusp by its 2048 link vertices; it does not count that cusp as an additional finite vertex. For

$$H(t) = (t-1)^{12} + \sum_{k=0}^{11} b_k (t-1)^k,$$

Dehn–Sommerville gives $H(t) = t^{12}H(1/t)$. Comparing coefficients and solving the six independent linear relations gives

$$\begin{aligned} f &= a_6 - 5a_7 + 14a_8 - 28a_9 + 42a_{10} - 42a_{11} - 462c, \\ a_2 &= 15a_6 - 74a_7 + 206a_8 - 411a_9 + 616a_{10} - 616a_{11} - 2684c. \end{aligned} \tag{14}$$

These two displayed coefficient rows can also be checked directly by substitution into the coefficient equations for H; there is no feasibility or enumeration premise. They give the identity

$$66f + 11264\lambda c - \frac{22}{5}a_2 = \frac{-22a_7 + 88a_8 - 198a_9 + 308a_{10} - 308a_{11}}{5}. \tag{15}$$

Every eight-face has at least 82 facets. Since each seven-face lies in five eight-faces, $5a_7 \geq 82a_8$. Similarly each ten-polytope has at least eleven facets and each nine-face lies in three ten-faces, so $3a_9 \geq 11a_{10}$. Thus

$$-22a_7 + 88a_8 \leq -\frac{1364}{5}a_8 < 0, \qquad -198a_9 + 308a_{10} \leq -418a_{10} < 0.$$

The last term in (15) is negative as well. This proves (13), with strictness.

7 Small three-faces and an exception budget

For a right-angled three-polyhedron write F, s, p for its polygon facets, ideal vertices and finite vertices, and put $\nu = F + s$. Finite and ideal vertex degrees are three and four. Euler gives

$$p + 2s = 2F - 4, \qquad p = 4F - 2\nu - 4. \tag{16}$$

A polygon with u finite and v ideal vertices has area $\frac{\pi}{2}(u+2v-4) > 0$, so the integer $u+2v-5$ is nonnegative. Summing over the polygons yields

$$\sum(u + 2v - 5) = F + 2s - 12 \geq 0. \tag{17}$$

In particular $p + 6s \geq 20$. If $F \leq 5$, Euler would give $p + 2s \leq 6$, $s \leq 3$ and $p + 6s \leq 18$, a contradiction. Thus $F \geq 6$ and $\nu \geq 9$.

Lemma 7.1. *Equality $F = 6$ or $\nu = 9$ occurs only for P_3.*

Proof. At $F = 6$, equation (17) gives $s \geq 3$. There are $24 - 2s$ polygon corners. At least $2s$ of the six polygons must therefore be triangles, so $s \leq 3$. Hence $s = 3, p = 2$, and all six facets are triangles. A positive-area right-angled triangle has at most one finite vertex. The finite polygon-corner count $3p = 6$ then forces each triangle to have exactly one finite vertex. At either finite vertex all three pairs are bad, and Lemma 2.2 identifies the whole polyhedron. Finally $F \geq 6$ and $F + 2s \geq 12$ make $F + s = 9$ possible only at $F = 6, s = 3$. □

7.1 Two triangles attached across their ideal edge

Lemma 7.2. *If two bad triangular facets of a right-angled three-polyhedron share their ideal–ideal edge, the whole polyhedron is P_3.*

Proof. Scale the shared future null endpoints u, v so that the first finite apex q_1 has products -1 with both. Its right angle gives $\langle u, v\rangle = -1$. Let n_1, n_2 be the orthogonal unit outward normals of the two triangles. They are perpendicular to u, v. The apices have the forms

$$q_1 = u + v - n_2, \qquad q_2 = tu + t^{-1}v - n_1, \quad t > 0.$$

Indeed the right angle at each apex makes the product of its two null coefficients one; unit timelike norm then makes the normal coefficient ± 1, with the negative sign fixed by orientation. The other four actual side normals are

$$u - n_2, \quad v - n_2, \quad tu - n_1, \quad t^{-1}v - n_1.$$

Two cross products are $-t, -t^{-1}$. Lemma 2.1 therefore forces $t = 1$. The third ideal vertex $w = u + v - n_1 - n_2$ is future null and is the sum $(u - n_1) + (v - n_2)$ of actual outward normals. Acuteness checks every other wall, and the selected walls also contain or accept w. The six known inequalities enclose exactly the finite-volume convex hull of q_1, q_2, u, v, w. All five vertices are feasible, so convexity identifies the whole polyhedron. □

7.2 The zero-excess eight-facet type

Call a three-polyhedron with $F = 8, s = 2$ a Q_8 type; metric uniqueness is unnecessary. Euler gives $p = 8$, and (17) says every polygon has zero excess. A bad triangle would share its ideal–ideal edge with another zero-excess polygon containing both ideal endpoints, hence another bad triangle. Lemma 7.2 would give P_3, a contradiction. There are no triangles. Its eight polygons have 32 corners in total, so every one is a $(u, v) = (3, 1)$ quadrilateral.

At a finite Q_8 vertex there is at most one edge with ideal other endpoint: two such directions would span a polygon containing two ideal vertices, impossible for its quadrilaterals. No edge joins its two cusps for the same reason. Their degree-four links supply eight finite–ideal edges; the eight finite vertices have at most one each, so each has exactly one.

7.3 Every finite corner has many exceptional three-faces

Call a three-face *exceptional* when it is neither P_3 nor a Q_8 type. At an ambient finite vertex q, let k be the number of its twelve directions whose edges end at ideal vertices, and let $t(q)$ be the number of triangles in G_q. Exactly $t(q)$ of its incident three-faces are P_3. A Q_8 can use only a triple with exactly one of the k ideal-ending directions, so the exceptional incidence count obeys

$$X(q) \geq 220 - t(q) - k\binom{12-k}{2}.$$

Every bad-graph triangle uses three ideal-ending directions, hence $t(q) \leq \binom{k}{3}$. For $k = 0, \ldots, 12$ one has

$$14\left(220 - \binom{k}{3}\right) - 27k\binom{12-k}{2} \geq 0.$$

The complete successive integer values are

$$3080, 1595, 650, 150, 0, 105, 370, 700, 1000, 1175, 1130, 770, 0.$$

It follows, with $\gamma = 13/27$, that

$$X(q) \geq \gamma(220 - t(q)), \qquad X := \sum_q X(q) \geq \gamma(220f - 2B). \tag{18}$$

This counts actual triples and ideal edge endpoints; no symmetrization of arbitrary corner graphs is used.

Lemma 7.3. *Every exceptional three-face satisfies*

$$p \leq 22(\nu - 10) + 4(8 - F). \tag{19}$$

Proof. Subtracting $p = 4F - 2\nu - 4$ from the right side gives $24\nu - 184 - 8F$. At $\nu = 10$, equation (17) gives $F \leq 8$. The six-facet type is P_3, and $F = 8, s = 2$ is Q_8, both excluded. The remaining $F = 7, s = 3, p = 4$ gives equality. At $\nu = 11$, one has $F \leq 10$, so the difference is $80 - 8F \geq 0$. At $\nu \geq 12$, use $F \leq \nu$ to obtain $16\nu - 184 \geq 8$. The case $\nu = 9$ is only P_3 by Lemma 7.1. □

8 Global incidence elimination and the contradiction

Define a nonnegative excess and a signed facet defect by

$$\mathcal{E} = \sum_{\text{non-}P_3} (\nu - 10), \qquad \mathcal{R} = \sum_{\text{non-}P_3} (8 - F).$$

The second quantity need not be nonnegative and is not a face count. Both contributions are zero for Q_8, so summing Lemma 7.3 over exceptional faces gives

$$X \leq 22\mathcal{E} + 4\mathcal{R}. \tag{20}$$

Each polygon is contained in ten three-faces. An ambient cusp belongs to $2^9\binom{11}{2} = 28160$ three-faces, by its cubical link. Hence the exact coface identities are

$$\mathcal{E} = B - 10a_3 + 10a_2 + 28160c, \qquad \mathcal{R} = 8a_3 - 10a_2 - 2B. \tag{21}$$

For example summing $\nu = F + s$ over *all* three-faces gives $10a_2 + 28160c$; removing the P_3 contribution to $\nu - 10$ adds B. Removing its $8 - F = 2$ contribution subtracts $2B$ in the other identity.

We also need the finite/cusp Euler incidence identity

$$10a_2 = 2a_3 + 110f + 28160c, \qquad a_3 = 5a_2 - 55f - 14080c. \tag{22}$$

To derive it, sum $F = 2 + p/2 + s$ from (16) over all three-faces. Each finite vertex belongs to 220 three-faces, each cusp to 28160, and the left side is again $10a_2$.

Combining (18)–(22) gives

$$101B - 5130a_2 + 69080f + 22049280c \geq 0. \tag{23}$$

For an explicit coefficient check, before substituting (22) and dividing by four the expression is

$$404B - 5076a_3 + 4860a_2 - 2860f + 16727040c \geq 0.$$

Using the strict weighted row (13) in (23) yields

$$101B > 7870f - 267300c. \tag{24}$$

Independently, Lemma 7.1 says $\nu \geq 10$ for every non-P_3 three-face, whereas P_3 has $\nu = 9$. Thus $\mathcal{E} \geq 0$ in (21); equivalently

$$B \geq 10a_3 - 10a_2 - 28160c = 40a_2 - 550f - 168960c.$$

The same strict weighted row gives

$$B > 50f + 880c. \tag{25}$$

Multiply (24) by 880 and (25) by 267300. Both multipliers are positive; the cusp terms cancel:

$$356180B > 20290600f, \qquad 2B > \frac{184460}{1619}f.$$

Strictness survives, and B, f are precisely the actual opposition face and finite-vertex counts used in Proposition 5.1. But

$$184460 \cdot 12500 - 1420561 \cdot 1619 = 5861741 > 0.$$

Since $f > 0$, this strict lower bound contradicts (10). There is no mixed right-angled twelve-polytope. Dufour's earlier ceiling twelve proves Theorem 1.1; the smaller compact and entirely ideal ceilings prove Corollary 1.2.

Proof and verification record

The proof uses actual hyperbolic normal geometry and finite incidence identities, with no arithmetic restriction, facet census or numerical optimizer. The full new scalar obligation is the polynomial expansion and interval certificate in Lemma 4.1, the rational moment substitution and the final integer comparison. A standalone standard-library exact checker has passed generation and a separate read-only replay. Its source and compact certificate are `scripts/check_right_angled_mixed_positive_part_opposition_cap_20261019.py` and `results/right_angled_mixed_positive_part_opposition_cap_research_20261019.json`. The earlier DS/moment/coupling checkers retain separate evidence; none replaces its geometric proof.

The module `formal/CoxeterProofs/RightAngledMixed11.lean` has compiled and passed a separate printed-axiom audit. It uses only the standard library. Its final count contradiction is conditional on the real transport/majorant upper bound, the polygon/three-face lower interfaces and identification of actual counts. These geometric premises are proved above, not formalized inside that module. The printed axioms are only `propext` and `Quot.sound`; the exact comparison is axiom-free, and there is no `sorryAx`. The manuscript's universal geometry has independent written reviews; this is not a claim of end-to-end geometric Lean verification or external refereeing. Typesetting and exact coefficient replay are distinct from the written proof. No example attaining eleven and no exclusion of dimension eleven is asserted.

References

[1] G. Dufour, *Notes on right-angled Coxeter polyhedra in hyperbolic spaces*, Geometriae Dedicata **147** (2010),277–282. https://doi.org/10.1007/s10711-009-9454-2. The published bound twelve is used; no equality classification is assumed.

[2] L. Potyagailo and E. Vinberg, *On right-angled reflection groups in hyperbolic spaces*, Commentarii Mathematici Helvetici **80** (2005),63–73. https://ems.press/content/serial-article-files/43032.

[3] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology **12** (2012),1941–1960. https://msp.org/agt/2012/12-4/agt-v12-n4-p03-s.pdf.

[4] S. Alexandrov, *On ideal vertices of right-angled hyperbolic polyhedra*, 2023, Sections 2–6. https://arxiv.org/html/2303.09533v1.